Unified Disaster Response Platform — Full Feature Specification

*Gap-closed version. Every gap identified in the first pass is now folded in as a named feature and marked **[GAP-n]**.*

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0. System model

Surfaces

Surface	Users	Notes
Command Centre (web)	State / national controllers	Map-first, multi-area, always online
Responder App (mobile)	CPOC, agency leads, field responders, camp registrars, police	Offline-first, mesh-capable
Intake Console (web)	112 / control-room operators	Civilian report entry **[GAP-3]**

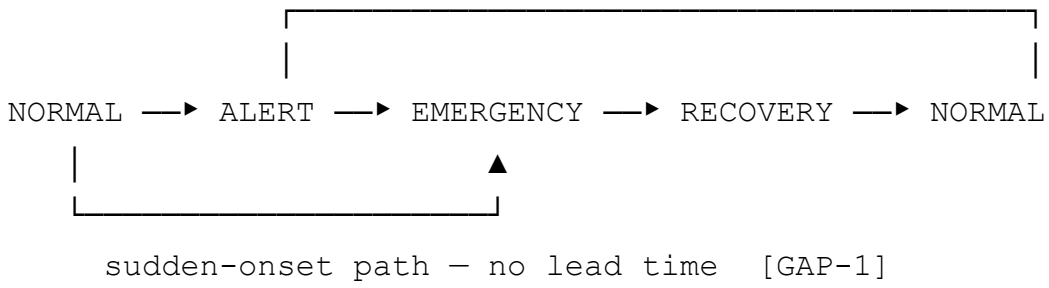
Actor hierarchy

```
Command Centre  (state / national)
      │  sets alert level · allocates inter-area resources · can force-activate
      ▼
CPOC  +  Deputy CPOC          ← one area each, deputy is mandatory  [GAP-2]
      │  assigns roles · approves activation · verifies · closes
      ▼
Agency Unit Leads             Fire · SDRF · NDRF · Police · Medical · Camp
      │
      ▼
Field Responders              individually authenticated, OTP
```

Supporting roles: **Camp Registrar** (Phase 3), **Police Officer** (missing persons), **Intake Operator** (civilian reports), **System Admin** (enrolment & config).

State machine

State is **scoped per area**, never global. District A can be in EMERGENCY while District B sits in ALERT.



Transition	Trigger	Authority
NORMAL → ALERT	Hazard feed crosses threshold	Command Centre (auto-proposed, human-confirmed)
ALERT → EMERGENCY	Field report approved	CPOC → Deputy → Command Centre (on timeout) **[GAP-2]**
NORMAL → EMERGENCY	Earthquake / collapse / industrial event	Auto-proposed by feed, confirmed by CPOC or CC **[GAP-1]**
EMERGENCY → RECOVERY	Active rescue phase declared over	CPOC, endorsed by Command Centre
RECOVERY → NORMAL	All incidents closed + camps wound down	Command Centre
Any → previous	False alarm / de-escalation	Command Centre only, reason mandatory

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1. Module map

#	Module	Serves	One-line purpose
M1	Identity & Access	All	Who is real, what they may do, offline
M2	Multi-Hazard Alert & Risk	P1	Turn hazard feeds into coloured areas + actions
M3	Activation & State Machine	P1→P2	Move an area between states, safely
M4	Mesh Comms & Sync	P2, P3	Move data with no towers, merge it without loss
M5	Incident Management	P2, P3	Report → triage → dedupe → assign → close
M6	Personnel, Tasking & Safety	P1–P3	Who is where, who is free, who is in trouble
M7	Civilian Intake	P2, P3	Public reports enter without breaking the closed system

#	Module	Serves	One-line purpose
M8	Evidence & Verification	P2, P3	Trustworthy photos, human verification
M9	People Registry	P3	Safe list, missing list, matching
M10	Command Centre Ops & Analytics	All	The bird's-eye view and the replay
M11	Audit & Accountability	All	Immutable record of every decision
M12	Admin & Configuration	All	Areas, agencies, thresholds, content

M5 is the spine. It is alive in Phase 2 (open/report mode) and Phase 3 (worklist/close mode) — same entity, different lens.

M1 · Identity & Access

Core

- OTP login against a **pre-registered responder database** — no public signup path exists.
- Personnel record: name, service number, agency, rank, area, capability tags, phone, device.
- Role-Based Access Control resolved as `(agency × rank × area × current state)`. A Fire Force sub-officer sees different actions in ALERT than in EMERGENCY.
- **Device binding** — one active device per responder; re-binding requires CPOC or admin approval. Stops credential sharing.

Offline authentication *(critical, easy to forget)*

- On successful online login the server issues a **signed session token with an expiry window** (e.g. 72 h) cached on the device.
- In mesh mode, peers verify each other using these tokens against a **locally cached roster of the area's personnel**, synced during ALERT. No server round-trip needed.
- A responder who never logged in before the towers fell cannot join the mesh — accepted trade-off; roster pre-sync during Phase 1 is what prevents it.

Continuity [GAP-2]

- **Deputy CPOC is mandatory** — an area cannot enter ALERT without one assigned.
- **Succession chain**: CPOC → Deputy → senior-most agency lead present → Command Centre.
- **Heartbeat**: if the CPOC device is unreachable for *N* minutes, authority auto-transfers to the Deputy with a banner notification to everyone in the area. Reversible on return.
- Explicit **handover** action for shift changes, with a handover note.

M2 · Multi-Hazard Alert & Risk [GAP-1]

Hazard ingestion adapters — one adapter interface, many sources:

Hazard	Source	Lead time
Rain / cyclone / heat	IMD	Hours–days
River flood	CWC	Hours–days
Cyclone / tsunami / storm surge	INCOIS	Hours
Earthquake	NCS	**None — sudden onset**
Landslide	GSI + rainfall correlation	Hours
Industrial / chemical / fire	Manual entry by Command Centre	None
Dam release / breach	State irrigation dept + manual	Minutes–hours

- Adapters normalise everything into a common `HazardSignal { type, area, severity, confidence, valid_from, valid_to, source }`.
- **Manual override**: Command Centre can raise or lower any area's level regardless of feed, with a mandatory reason.
- **Sudden-onset behaviour**: hazards with no lead time skip ALERT entirely and propose EMERGENCY directly (M3).

Risk & measures library

- Content keyed by `(hazard type × alert level × area profile)` — coastal vs hill vs urban vs industrial.
- Each entry produces: **likely risks**, **measures to carry out**, **equipment to stage**, **pre-positioning suggestions**.
- Editable by admins (M12) so states can tune language and local practice — this is what makes it deployable beyond a demo.

Map layers

- Alert colour per area (Red / Orange / Yellow / Green).
- Hazard-specific overlays: flood inundation model, cyclone track cone, landslide-susceptible slopes, seismic intensity shake map.
- **Static vulnerability layers** pre-loaded offline: hospitals, schools, shelters, camp sites, helipads, bridges, power substations, chemical plants, elderly/disabled clusters if the state has that data.

M3 · Activation & State Machine

Standard activation (Phase 1 → 2)

1. Local unit files an **Emergency Activation Request**: photos, voice note, text, location, hazard type, first severity estimate.
2. CPOC reviews in a dedicated approval screen (evidence side by side with the risk map).
3. Approve → the entire area flips to EMERGENCY. Every app in the area changes mode automatically.

On flip to EMERGENCY, the app auto-reconfigures:

- Mesh radios on, discovery aggressive
- Location reporting interval drops (e.g. 5 min → 30 s)
- Emergency UI: large targets, high contrast, one-tap reporting
- Battery-saver profile engages (M4)
- Offline map tiles for the area force-load if not already cached

Escalation safeguards [GAP-2]

- **Approval timeout ladder**: unapproved after T_1 → Deputy CPOC notified; after T_2 → Command Centre may approve unilaterally; after T_3 → auto-approve with a flagged provisional status.
- **Multi-report auto-trigger**: K independent activation requests from different agencies in the same area within a time window → auto-escalate to provisional EMERGENCY pending confirmation. Prevents a single unreachable CPOC from blocking the whole system.
- **Command Centre force-activation** — always available, always logged.

De-escalation

- False alarm rollback with mandatory reason; incidents raised during the false window are retained and marked, not deleted.
- **Cascade handling**: an aftershock or second breach during RECOVERY re-enters EMERGENCY without losing Phase 3 state — camps and closed incidents persist.

M4 · Mesh Comms & Sync [GAP-6]

Transport

- BLE for discovery and low-bandwidth messaging; Wi-Fi Direct / Nearby Connections for bulk (photos, map deltas).
- **Multi-hop routing** with dynamic re-route as nodes move; each device is a relay.
- **Gateway auto-detection** — any node that finds internet (NDRF vehicle satellite terminal, a working tower at the edge, a drone pass) announces itself; the mesh drains its queue through it.
- **Data-mule / store-and-forward** — queued payloads carried physically by a responder or drone moving between clusters.

Priority queue — bandwidth is the scarcest resource, so ordering is a feature, not an implementation detail:

```
P0  SOS · man-down · backup request
P1  State changes · tasking · assignment
P2  New incident reports (text + metadata)
P3  Location beacons
P4  Media (thumbnails first, full-resolution last)
```

Conflict-free merge [GAP-6]

- The incident log is **append-only**. Nothing is edited in place — every action is an `IncidentEvent`.
- Each device stamps events with `(device_id, monotonic_seq, vector_clock, wall_clock)`.
- Merge on reconnect is a deterministic replay by causal order, so two responders working the same incident offline both keep their contributions — no overwrite, no lost report.
- **Only CPOC's `CLOSE` event is authoritative.** Field responders can only mark `RESOLVED_PENDING_VERIFICATION`. This makes closure a single-writer operation and removes the hardest class of conflict entirely.
- Concurrent conflicting metadata edits (two different severities) surface to CPOC as a **flagged divergence** rather than silently picking a winner.

Efficiency

- Ultra-low-bitrate voice codec for push-to-talk.
- On-device object detection sending structured JSON alerts instead of video.
- Delta-sync for map annotations, not full state.
- **Battery governor**: mesh scan interval adapts to remaining battery; below a threshold the device becomes a leaf node (receives, stops relaying) to preserve enough charge for its own SOS.

M5 · Incident Management

Incident lifecycle

```
REPORTED → TRIAGED → ASSIGNED → IN_PROGRESS →
RESOLVED_PENDING_VERIFICATION → CLOSED
                                                  │
                          ├──▸ DUPLICATE (linked to parent)  │
                          ├──▸ FALSE / NOT_FOUND             │
```

```
                                  └──► REOPENED ◄──────────────────────┘
(verification rejected)
```

Reporting

- **One-tap structured reporting** — 8–12 icon buttons (collapse, trapped persons, flooding, fire, road blocked, medical, body recovered, landslide, power line down, gas leak, water/food need, other). Their value is auto-filling the shared map fast under stress.
- Free-text description with **multilingual entry + translation** so any agency can read any report.
- Voice note attachment.
- Auto-attached: GPS, timestamp, reporter identity, agency, device.
- Optional photo via the locked in-app camera (M8).

Triage & severity [GAP-5]

Severity is captured **at report time**, mandatory, single tap:

Level	Meaning	Target response
P0	Life at immediate risk — trapped, drowning, active fire with occupants	Immediate
P1	Serious but not immediate — injured stable, isolated group needing extraction	Hours
P2	Infrastructure / access — road blocked, power line down, bridge damaged	Same day
P3	Logistics / welfare — supply need, sanitation, damage survey	Post-rescue

- CPOC can re-triage; every change is an audited event.
- **Ageing indicator** — a P0 open beyond its target visually escalates on the map. This is what stops the worklist from becoming a flat pile.
- Priority ordering: `severity → age → proximity of an available unit`.

Duplicate detection [GAP-4]

Concrete rules, not vibes:

- **Auto-link candidate** when: same incident type **AND** within **100 m AND** within **30 min**.
- **Soft-flag candidate** when: same type AND within **300 m** AND within **2 h** — shown to CPOC as "possible duplicate", not auto-linked.
- Radius is **type-dependent**: a building collapse uses 50 m, area flooding uses 500 m. Configurable per type in M12.

- Linking creates a **parent incident with children**; all child reports, photos, and reporters are preserved and visible under the parent. Nothing is deleted.
- **CPOC can link and unlink manually** at any time — the automatic rule only proposes.
- **Provenance is kept**: "4 reports from 3 agencies" is displayed on the parent, because multiple independent reports are itself a signal of severity.

On the map

- Every incident is a map point, colour = severity, shape/icon = type, ring = status.
- Cluster view at low zoom with counts.
- Filters by agency, type, severity, status, age, assigned unit.
- **Offline vector base map** so this all works with no connectivity.

M6 · Personnel, Tasking & Safety

Roster & availability

- Live roster per area with status: `AVAILABLE` · `EN_ROUTE` · `ENGAGED` · `RESTING` · `OFF_DUTY` · `UNREACHABLE`.
- **Capability tags**: swift-water rescue, rope/height, K9, medical, cutting equipment, boat operator, drone pilot, diver.
- Shift and fatigue tracking — hours on task, mandatory rest flags.

Role assignment (Phase 1)

- CPOC allocates each agency its responsibility before impact; acknowledgement required, non-acknowledgement visible.

Task assignment (Phase 2–3)

- Assign incidents to units directly from the map.
- Unit accepts / declines with reason; decline re-routes.

Backup request routing

- One-tap backup request from any responder.
- Routing considers: **nearest → available → capability match → not already on a higher-priority task**.
- Acceptance timeout → auto re-route to next candidate; CPOC sees the whole chain.
- Over mesh, this is a P0 message — it hops until someone acknowledges.

Responder safety *(new — responder welfare was a documented gap in the underlying research)*

- **Check-in timer**: responders entering a hazardous task set a timer; missed check-in raises an alert to CPOC and nearby units.
- **Man-down detection**: no device movement for *N* minutes during an active task → automatic alert with last known position.
- **Personal SOS** — highest-priority message in the mesh, broadcast to every node in range regardless of agency.
- **Last-known-position** persists on the map when a responder goes offline, greyed with the timestamp — so nobody disappears silently.

Resources

- Vehicle, boat, and equipment inventory per unit, with location and status.
- Consumables tracking for Phase 3 (relief material stock at camps).

M7 · Civilian Intake [GAP-3]

The responder network stays closed. Civilian information enters through a controlled boundary instead.

Intake Console (control room / 112 operators)

- Operator enters a call as a `CivilianReport`: caller name, callback number, location (address → geocoded, or map pin), description, claimed severity.
- Reports land in a **staging queue**, visibly tagged `UNVERIFIED · CIVILIAN`, and are **not** pushed to responder devices until promoted.
- CPOC or a delegated verifier promotes a report into a real incident, optionally after a callback.
- Duplicate detection (M5) runs across civilian and responder reports together — a citizen call and a fire-force sighting of the same collapse link into one parent.

Optional public channels (design them now, ship them later)

- **SMS gateway** with a fixed short format for zero-data areas.
- Lightweight public web form, no login, heavily rate-limited.
- **Spam and flood control**: per-number rate limits, geographic plausibility check, automatic clustering of mass reports about the same event so 500 calls about one flooded street become one incident with a count of 500 — which is a *stronger* signal, not noise.

Trust model — every incident carries a `source` field: `RESPONDER · CIVILIAN_VERIFIED · CIVILIAN_UNVERIFIED · SENSOR · MEDIA`. Responders always see where information came from before acting on it.

M8 · Evidence & Verification

Locked capture

- Photo capture happens **inside the app only**. Gallery upload is disabled — this is the whole trust argument.
- Bound at capture time: GPS coordinates, GPS accuracy, timestamp (device + server-corrected on sync), device ID, responder ID, incident ID.
- Payload hashed and signed with the device key at capture; any later alteration breaks the hash.
- Photos taken offline queue with their **original** timestamp, not the sync time.

Verification workflow

- Field responder marks an incident `RESOLVED_PENDING_VERIFICATION` and attaches closure evidence.
- **Before/after pairing** — the original report photo and the resolution photo shown side by side to CPOC.
- **Geo-fence check** — automatic flag if the closure photo's coordinates fall outside a tolerance radius of the reported incident.
- CPOC verification queue: approve → `CLOSED`, map point turns green. Reject → `REOPENED` with a mandatory reason returned to the responder.
- Command Centre can spot-audit any closed incident and reopen it.

M9 · People Registry (Phase 3)

Two tracks that meet in a matching engine.

Track A — Safe list (camp registration)

- Camp Registrar checks in every person entering a relief camp: name, age, sex, contact number, origin village/ward, **family group ID**, arrival time, optional photo, special needs (medical, mobility, unaccompanied minor, pregnant).
- Family grouping so households register once and stay linked.
- Check-out recording when people leave, so the list stays accurate.
- Camp-level counts feed logistics: occupancy, food/water/bedding demand, medical needs.

Track B — Missing list (police)

- Police file missing-person reports in-app: name, age, sex, description, last known location, photo, reporting relative's contact.
- Case status: `OPEN` · `PROBABLE_MATCH` · `REUNITED` · `DECEASED` · `CLOSED_UNRESOLVED`.

Matching engine

- Fuzzy name matching tolerant of transliteration variants (a real problem across Tamil/Hindi/Bengali → Latin spellings).
- Scored on: name similarity + age band + origin location + family group overlap.
- Produces **ranked candidates for human confirmation** — never auto-confirms. A wrong automatic match on a missing child is unacceptable.
- On confirmation: both records update, the reporting relative is notified through the police officer.

Family enquiry

- Police, CPOC, and Command Centre can query the safe list by name/village/phone and answer a family in seconds instead of camp-by-camp.
- Cross-camp search — a person registered in Camp 3 is findable by someone asking at Camp 1.

Restricted records

- Casualty and deceased records handled as a **separate record type with tighter access** (police + Command Centre only), never displayed on the general map.

Privacy

- No public exposure of the safe list; access is role-scoped and every lookup is audited (M11).
- Retention policy: personal data purged or archived after the declared disaster period.

M10 · Command Centre Ops & Analytics

- **Multi-area overview**: every area with its state, open incident counts by severity, personnel deployed, resources committed.
- **Progress tracking**: open vs closed per area; percentage green; an area only turns green when every incident in it is closed.
- **Heat views**: incident density, unmet-need density, response-time outliers.
- **Resource allocation view** — where units are concentrated vs where POs are concentrated. This surfaces misallocation, which was one of the original problems being solved.
- **Inter-area resource requests** from CPOCs, with approve/deny and dispatch tracking.
- **Timeline replay** — scrub the whole event: how the map filled, when units moved, when things closed. Powerful in a demo and genuinely useful for training.
- **After-action report export** — incident counts, response times per severity, closure rates, agency contribution, camp populations over time, unresolved items.

M11 · Audit & Accountability

- Every consequential action is an immutable, append-only log entry: activation requests, approvals, escalations, timeouts, re-triage, assignments, links/unlinks, verifications, rejections, closures, reopens, registry lookups, permission changes.
- Entries record actor, role, timestamp (original, not sync time), area, and target entity.
- Offline actions carry their original timestamp and are marked as synced late — so the log reflects what actually happened when.
- **Read-only** for everyone including admins; exportable for enquiry commissions and after-action review.

M12 · Admin & Configuration

- **Area definition**: boundaries, area profile (coastal/hill/urban/industrial), assigned CPOC and Deputy.
- **Agency & personnel enrolment**: bulk import, verification, deactivation, device re-binding.
- **Content library**: risk and measures text per hazard × level × profile, in all supported languages.
- **Thresholds**: dedupe radius and time window per incident type, escalation timeouts $T_1/T_2/T_3$, multi-report auto-trigger K, severity response targets, check-in and man-down intervals.
- **Language packs**: add/remove, manage on-device size.
- **Map data**: offline tile packs and vulnerability layers per area, versioned.
- **Drill mode** — run the whole state machine in a sandbox for training without touching real records. Cheap to build, very persuasive to a judge.

Gap resolution summary

#	Gap	Resolved by
1	Phase 1 only worked for weather	M2 multi-hazard adapters + sudden-onset path in M3
2	CPOC single point of failure	Mandatory Deputy, succession chain, heartbeat transfer, escalation timeout ladder, multi-report auto-trigger (M1, M3)
3	No civilian channel	M7 Civilian Intake with staging queue, verification, source tagging, flood control
4	Dedupe was vague	M5 concrete type-dependent radius/time rules, parent-child linking, manual link/unlink, provenance preserved
5	No severity or priority	M5 P0–P3 triage at report time, ageing escalation, priority-ordered worklist
6	Mesh edit conflicts	M4 append-only event log, causal merge, single-writer closure, divergence flagging

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Data model sketch

```
Area(id, name, boundary, profile, cpoc_id, deputy_id, current_state,
alert_level)
HazardSignal(id, type, area_id, severity, confidence, valid_from,
valid_to, source)
User(id, name, service_no, agency, rank, area_id, capabilities[],
device_id, status)
Incident(id, parent_id, type, severity, status, lat, lng, area_id,
         source, created_at, created_by, assigned_unit_id, closed_at,
closed_by)
IncidentEvent(id, incident_id, event_type, actor_id, payload,
              device_id, seq, vector_clock, wall_clock, synced_at)
Media(id, incident_id, kind, lat, lng, accuracy, captured_at, device_id,
      captured_by, hash, signature)
Task(id, incident_id, unit_id, state, accepted_at, completed_at)
BackupRequest(id, requester_id, incident_id, routed_to[], accepted_by,
created_at)
CivilianReport(id, operator_id, caller_contact, description, location,
               status, promoted_incident_id)
Camp(id, name, area_id, capacity, occupancy, location)
Person(id, camp_id, name, age, sex, contact, origin, family_group_id,
       needs[], checked_in_at, checked_out_at)
MissingReport(id, filed_by, name, age, sex, description, last_seen, photo,
status)
MatchCandidate(missing_id, person_id, score, confirmed_by, confirmed_at)
AuditEntry(id, actor_id, role, action, target_type, target_id,
           original_ts, synced_ts, area_id)
```

Build order (dependency-first)

Since you're pre-building, build in dependency order rather than phase order — Phase 1 features are not the easiest ones.

1. **Foundation** — M12 area/agency/personnel config, M1 auth + RBAC, offline token scheme
2. **Spine** — M5 incident entity, event log, map, offline tiles
3. **Trust** — M8 locked camera + verification queue (this makes M5 closable)
4. **State** — M3 state machine + M2 alert levels (M2 can start with one hazard source and a manual override)

5. **Transport** — M4 mesh + priority queue + merge (build against the M5 event log from step 2, which is why the log comes first)
6. **Coordination** — M6 roster, tasking, backup routing, safety
7. **Phase 3** — M9 registry + matching
8. **Edges** — M7 civilian intake, M10 analytics, M11 audit views

Steps 2 and 3 together give you the demo loop that actually wins:
report with photo → CPOC approves → area red → map fills → unit resolves → geotagged proof → CPOC verifies → point turns green.
Everything else deepens that loop rather than replacing it.